

Experiment - 1

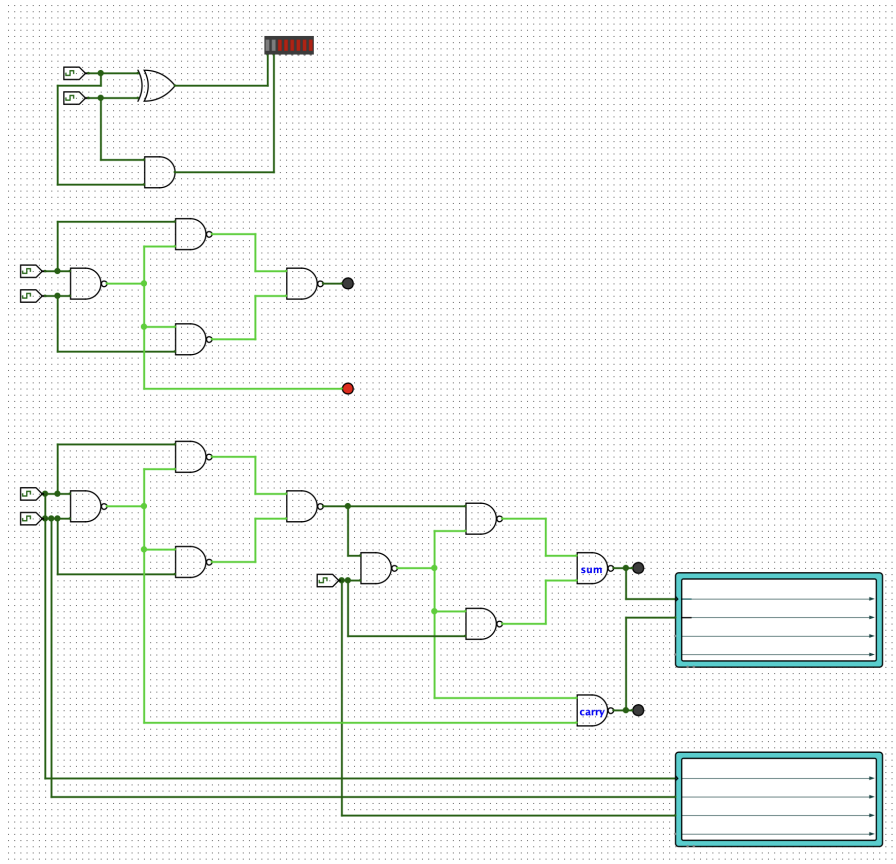
Aim

The objective of this experiment is to design and verify the operation of a Half Adder, Full Adder, and 4-bit Adder using basic logic gates in Logisim Evolution, with sum and carry outputs observed on output display components.

Procedure

1. Opened Logisim Evolution and created a new project file for the experiment.
2. Designed the Half Adder in the upper section of the canvas using one XOR gate for the Sum output and one AND gate for the Carry output, with two single-bit inputs.
3. Connected the Sum output of the Half Adder to an output display component at the top to observe the result.
4. Verified the Half Adder by applying all four input combinations (00, 01, 10, 11) and confirming correct Sum and Carry values.
5. Designed the first Full Adder in the middle section using two AND gates, two XOR gates, and one OR gate to process inputs A, B, and Cin.
6. Designed a second, more complete Full Adder in the lower section of the canvas, explicitly labeling the Sum and Carry output probes for clarity.
7. Connected the labeled Sum output to one display register and the Carry output to a separate display register on the right side of the circuit.
8. Tested the Full Adder by applying all eight possible input combinations of A, B, and Cin and verifying that the Sum and Carry outputs matched the expected truth table.
9. Extended the design concept to a 4-bit adder by cascading four such Full Adder stages, connecting Cout of each stage to Cin of the next.

10. Observed and recorded the outputs from both display components for various input combinations to confirm correct operation.



Results

The simulation verified correct operation of both the Half Adder and Full Adder for all input combinations. The labeled Sum and Carry outputs on the display components confirmed expected logic values. The cascaded structure demonstrated accurate binary addition with carry propagation working correctly across stages.